REDA ELMOUNTASSIR

Philadelphia, PA | redaelmountassir0@gmail.com | (267) 231-2928 | linkedin.com/in/reda-elmountassir | redaos.netlify.app

Mechanical Engineering student (Penn) with a hands-on background in CAD, mechatronics, and fabrication, plus full-stack software development experience passionate in social work and entrepreneurship.

EDUCATION

UNIVERSITY OF PENNSYLVANIA | Bachelor of Science in Engineering, Mechanical Engineering (GPA: 3.78) *May 2029*

COMMUNITY COLLEGE OF PHILADELPHIA | Associate's Degree in Business General (GPA: 4.0) *Apr 2025*

Honors & Awards: Anne and Andrew Abel Scholarship and Writing Award

MAST COMMUNITY CHARTER SCHOOL | High School Diploma (GPA: 4.0) *Jun 2025*

Honors & Awards: Valedictorian, VP of National Honor Society, President of Robotics Club, VP of Muslim Students Association

SKILLS

MECHANICAL | CAD, Arduino, Soldering, Circuitry

TOOLS | React, Next.js, Gatsby, Astro, Tailwind, GraphQL, Zustand, NumPy, Docker, SQL, PyTorch, Blender, Adobe Suite

INTERESTS | Virtual Reality, Game Design, Soccer, Darija (Arabic)

PROJECTS

BALLISTA *May 2026*

- Engineered a marble-launching mechanism in SolidWorks to fit within a sq. ft. footprint and launch marbles over a yard in distance
- Fabricated components with a 3-person team via laser-cut acrylic and MDF; 3D-printed the carriage for the ball

DALEK REPLICA *Feb 2020*

- Diagnosed and repaired the 5ft, battery-powered replica of a DALEK, built on dual Raspberry Pi circuits for head and body movement
- Designed remote-control circuitry and housing in CAD; programmed controller logic in Python/C++, reducing packet size and handling data-loss and quality-of-life edge cases

BATTLE BOT *Jan 2016*

- Repaired and optimized a competitive battle bot with the Franklin Institute team, resoldering components and replacing worn parts
- Contributed to first-place finishes in 2 of 2 competitions entered

ADDITIONAL PROJECTS | Portfolio Site (Gatsby/GraphQL, 50%+ faster load times), Cloud Renderer (real-time volumetric clouds, Unity/HLSL), Draft Studios Site (Next.js)

PROFESSIONAL EXPERIENCE

PRODUCT DEVELOPER

Zero Vector Ventures | Philadelphia, PA *Feb 2025 – Aug 2026*

- Evaluated 5+ problems/products monthly through a multi-stage validation pipeline with the goal of acquisition or IPO
- Engineered internal tooling for a Kanban-based workflow, incorporating automated AI-driven review steps

FULL-STACK DEVELOPER / DEV LEAD

Coded by: | Philadelphia, PA *May 2021 – Feb 2025*

- Wrote comprehensive documentation for internal Docker/Kubernetes workflows
- Trained and mentored 4 generations of high-school and college interns in full-stack development; guided pre-seed-stage solution pitches

RESEARCHER & DEVELOPER

Franklin Institute of Philadelphia | Philadelphia, PA *Oct 2015 – Jul 2023*

- Designed, constructed, and repaired 4+ mechanical showcases for the museum floor
- Taught dozens of students on introductory programming concepts (Git, Python, UDP/TCP) applied in exhibit construction

CERTIFICATIONS

SOLIDWORKS DESIGN ASSOCIATE (CSWA) *May 2026*